

MolarMind

As-built architecture for an offline-first chairside agent that turns hygienist speech into a periodontal chart, patient-plain language, and mocked Open Dental write-back.

System

MolarMind (working title: Chairside Agent)

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Audience

Product, engineering, and clinic IT

Document

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As-built, demo grade, pre-production

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Internal architecture pack

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This pack describes the running system in the chairside-agent repository, not the abandoned FastAPI plus WebSocket sketch in the original BUILD note. Where those two diverge, the browser-only design is the decision.

1. Decision summary

MolarMind is a **single-origin browser application**. Capture, parse, chart, translate, and PMS preview all run in the clinician tab. A second tab on the same origin is the patient surface. They share state through BroadcastChannel, not a server.

Speech never leaves the operatory on the critical path. Whisper runs in a Web Worker via Transformers.js. Charting is a deterministic parser. Low-confidence transcripts appear in Heard and do not write the exam or the Open Dental mapper.

Practice-management integration is contracted against the public Open Dental REST API (perioexams, periomeasures, procedurelogs). Today those calls are **built and displayed, not sent**. That keeps the demo honest about the write-back shape without taking a network or HIPAA dependency.

The driving principle is **stage safety and reversibility**: boring local tech, a scripted rehearsal path, and a PMS adapter that can later point at a real clinic without rewriting the charting core.

2. Context and constraints

What is being built

A hands-free agent for the dental operator. One voice stream serves two people: the hygienist gets a live perio grid and a detailed report; the patient gets plain-language captions, a jaw map, and March-to-today comparison. The business case is engagement, not another silent ceiling-tile cleaning.

The seeded patient is Andrew. Last visit is 12 March 2026. Today's exam starts empty and fills as speech (or the rehearsal script) arrives. Watch and needs-care teeth, plus a crown on #30 and fillings on #15 and #19, are the demo story.

Upstream constraints

- Demo-grade product. It must not fail on stage. Offline is a feature, not a fallback.
- No large language model on the charting path. Determinism beats warmer copy.
- No real clinic credentials, no real PHI, no live PMS writes in this build.
- Host, port, routes, Whisper model, and Open Dental keys come from environment files, not hardcoded loopback URLs in application code.
- The original BUILD note assumed Python FastAPI, faster-whisper, and a mock HTTP server. The shipped system dropped the process boundary.

Non-goals

- HIPAA program, clinic SSO, multi-tenant hosting, or audit export.
- Live Open Dental or Dentrrix writes.
- Imaging, cameras, insurance, scheduling, or full treatment planning.
- Photoreal 32-tooth 3D. The patient surface uses photographed arches plus focused meaning, not a full-mouth mesh.

3. Topology

The system is a **browser monolith with two peer windows**. Vite serves static assets. There is no application server, no WebSocket hub, and no database process.

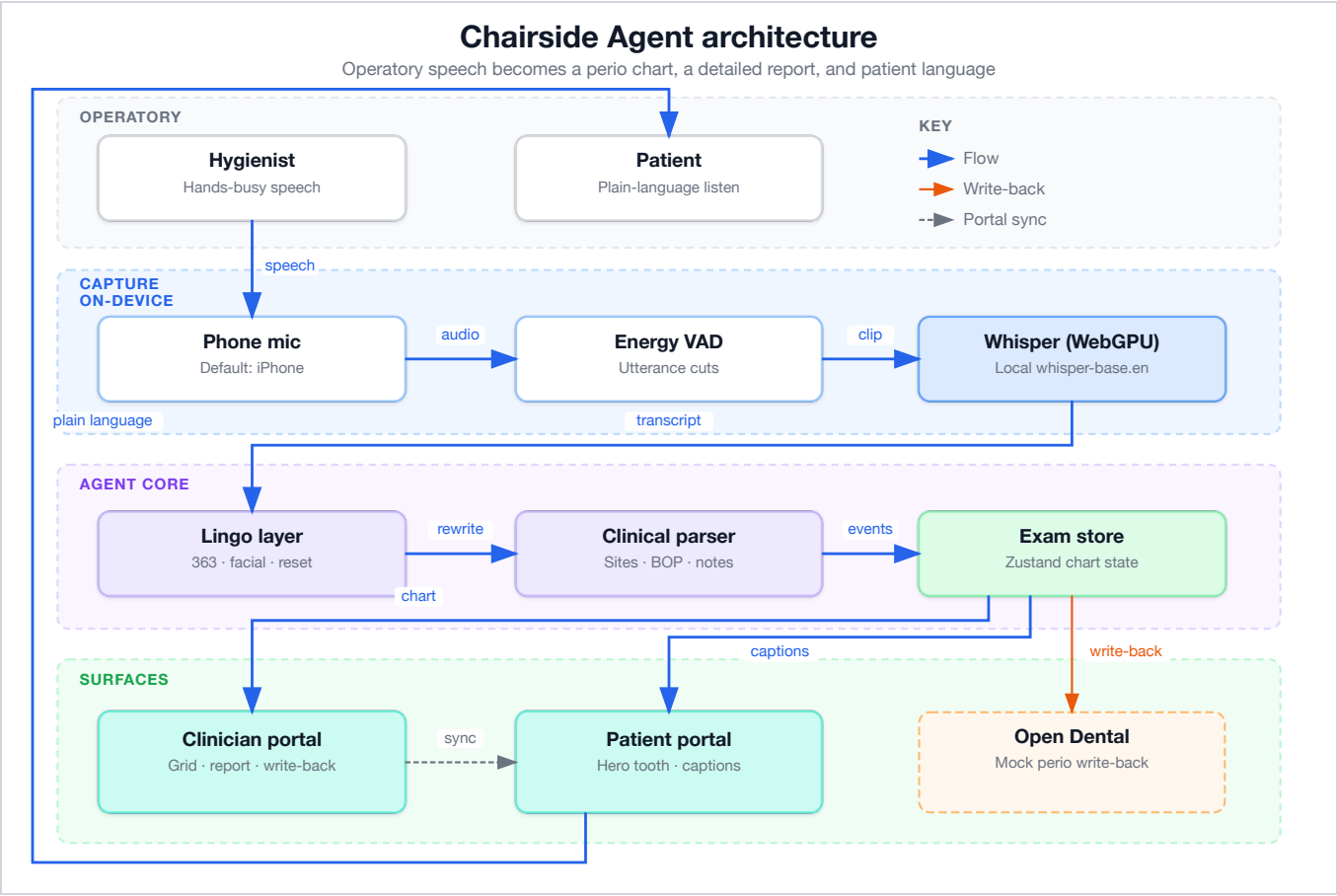


Figure 1. System context. Speech stays in the operatory. Chart events fan out to clinician, patient, and the Open Dental mapper.

Runtime boxes

Box	Where it runs	Job
Clinician window	Browser tab, clinician route	Mic, VAD, Whisper host, parser, store, grid, report, write-back panel, rehearsal script
Patient window	Browser tab, patient route, same origin	Jaw map, captions, meaning panel, March-to-today slider. Consumes the same Zustand store when it is the same tab; uses the bus when it is a second tab listening to chart-events
Architecture window	Browser tab, architecture route	Shows the public SVG. This PDF is the detailed companion.

Whisper worker	Web Worker in the clinician tab	Loads whisper-base.en (q8, WebGPU) and transcribes one utterance at a time
Open Dental	Not contacted	Contract only. Mapper emits method, path, headers, body, and a mocked 201 or 200

Why not the original three-process design

Choice: one Vite app, no FastAPI, no WebSocket.

Rejected: Python STT service plus browser clients.

Why: one fewer process to crash on stage, one fewer port to bind, no wifi, no Python toolchain on the demo laptop. The cost is that Whisper download and WebGPU quality now sit in the browser.

4. Component decisions

4.1 Application runtime

Choice: Vite 8 + React 19 + TypeScript + React Router.

Alternatives: Next.js (rejected: server runtime we do not want); Electron (rejected: extra packaging for a two-window laptop demo); Python UI (rejected: patient visuals are web-native). **Tradeoff:** COOP and COEP headers are required so Transformers.js can use the worker and WebGPU path.

Routes are configurable: clinician, patient, and architecture paths come from `VITE_*` env. Default host and port also come from env. Application code must not hardcode loopback URLs.

4.2 Capture and speech-to-text

Choice: `getUserMedia` + energy VAD + Transformers.js Whisper in a worker, model `onnx-community/whisper-base.en`, dtype `q8`, device WebGPU.

Alternatives: `faster-whisper` in Python (original spec); cloud STT (rejected: wifi and PHI); `whisper-small` in-browser (heavier, only if base misses terms). **Tradeoff:** first-load model fetch can hit the Hugging Face CDN unless the model is cached. After cache, the path is local. Demo laptops should warm the model once before the room goes dark.

Setting	Value	Why
VAD threshold	RMS 0.012	Cuts chairside water and suction better than a hair-trigger
Silence to cut	550 ms	Perio callouts are short phrases
Minimum speech	700 ms	Drops chair clatter
Transcribe timeout	20 s	Hard stop so a stuck worker cannot hang the exam
Resample	16 kHz	Whisper contract
Preferred mic	iPhone Continuity, then Shokz, then headset	Operatory: phone on a tray or bone-conduction mic, not the laptop fan

Device classification is label-based (includes, not regular expressions). Only one tab should Listen. Continuity mics are exclusive and mute when the phone locks.

4.3 Lingo and parser

Choice: two-pass deterministic pipeline. Lingo rewrites chairside slang and detects commands. The parser plus clinical extractor emit `ChartEvent` objects.

Rejected: LLM slot filling on the live path (nondeterministic, latency, stage risk); a single giant regex (fragile, and this repo forbids regex-led parsing). **Tradeoff:** new slang needs a lexicon entry. That is acceptable. Charting bugs are worse than a missed synonym.

Parser rules that the architecture treats as load-bearing:

- Tooth navigation sets context ("tooth fourteen", "number 30").
- Side words map facial or palatal onto buccal or lingual site triplets.
- Triplets and named sites write PD. "Bleeding" sets BOP. Recession, mobility, and furcation are first-class fields.
- Corrections replace the last write. They must keep working.
- Exam-level negatives ("no bleeding") land as exam notes, not tooth PD.
- `confidence`: "low" never mutates pocket values in `applyEvent` and never enters the PMS mapper.
- Table-driven tests (51 utterance cases at last count) are part of the contract, not optional coverage.

4.4 Exam store

Choice: Zustand in-memory store as the system of record for the visit.

Rejected: disk database, remote session server, `localStorage` as source of truth (accidental PHI residue on a shared laptop). **Tradeoff:** refresh loses today's exam. Reset is explicit. Last visit reloads from Andrew's seed JSON.

The store holds last visit, current exam, Heard, captions, summary, focus tooth, timeline (0 to 1), and the Open Dental call list. `applyChartEvent` applies domain rules, then asks the PMS mapper for calls, then publishes UI state. `resetExam` clears the visit and the mock Open Dental session counters.

4.5 Clinician and patient surfaces

Choice: two routes, two physical displays. Clinician is a dense dark grid. Patient is a calm jaw-photo map plus meaning copy.

Rejected: one window with a split pane (too small on a laptop plus projector); a native iPad app (out of scope). **Tradeoff:** the original 3D hero tooth is no longer the center of the patient story. Photographed arches plus restoration icons are clearer in a two-day demo and still read from 10 feet.

Status colors are a product contract: green at or under 3 mm, amber at 4 mm, red at 5 mm and above. History teeth without a today reading show March color until measured.

4.6 Practice management adapter

Choice: Open Dental public REST as the PMS contract. In-browser mapper builds real requests.

Alternatives: Dentrix / Eaglesoft (partner-gated APIs, no public perio write spec we can mock honestly); a fictional `POST /mock/opendental/perio` blob (rejected: it does not teach the real write shape); live HTTPS from the demo laptop (rejected: keys, network, and accidental writes). **Tradeoff:** Open Dental is not the majority US PMS. It is the one with documented perioexams and periomeasures. The mapper is isolated so a later Dentrix adapter can sit beside it.

Base URL and demo keys are env: `VITE_OPENDENTAL_API_BASE`, developer and customer keys, `PatNum`, `ProvNum`. Defaults match Open Dental's own examples (`PatNum 20`, `ProvNum 3`, API host `https://api.opendental.com/api/v1`).

4.7 Rehearsal and script

Choice: hygienist cleaning script plus `?script=demo1` spacebar advance. Same parser, same store, same write-back.

Rejected: recorded audio playback as the only fallback (rooms kill speakers; spacebar never does).

5. Data architecture

Stores

Store	Purpose	Lifetime
src/data/andrew.json	Seed last visit and identity for the demo patient	Repo, versioned. Fictional.
Zustand useExamStore	Today's exam, UI, write-back log	Tab memory. Gone on refresh or Reset chart.
Open Dental session module	PerioExamNum, measure ids, procedure dedupe	Module memory. Reset with the exam.
Browser cache for Whisper	ONNX weights	Origin cache. Not PHI.
BroadcastChannel chairside-events	Fan-out of chart events to same-origin tabs	In flight only. Not durable.

Conceptual exam model

```
Exam
  patientId, patientName, date
  notes[]                exam-level (including negatives)
  teeth[1..32]
    sites[MB,B,DB,ML,L,DL]
    pd?, bop?, rec?
    mobility?, furcation?
    notes[], missing?

ChartEvent
  kind: reading | flag | navigation | summary_request
       | reset_tooth | reset_all | set_name
  tooth?, teeth[], side?, sites[], readings[]
  bopSites[], rec?, recSites[]
  mobility?, furcation?
  note?, notes[], examNotes[]
  raw, confidence: high | low
```

Consistency

Single writer: the clinician tab's store after a parsed high-confidence event. Patient UI is a reader. There is no conflict resolver because there is no multi-writer clinic graph.

PMS session consistency: first write of a visit creates one mocked PerioExam. Later surfaces on the same tooth PUT the existing PerioMeasure. That matches Open Dental's "one SequenceType per tooth per exam" rule.

Retention and regional strategy

Demo retention is "until Reset or reload." There is no backup and no region. Production would invert this: durable visit records in a clinic-controlled store, regional residency matching the PMS, and no Whisper weights mixed with PHI.

6. Integration patterns

Internal integration is in-process function calls plus one same-origin bus. External integration is a typed mapper that pretends to be HTTPS.

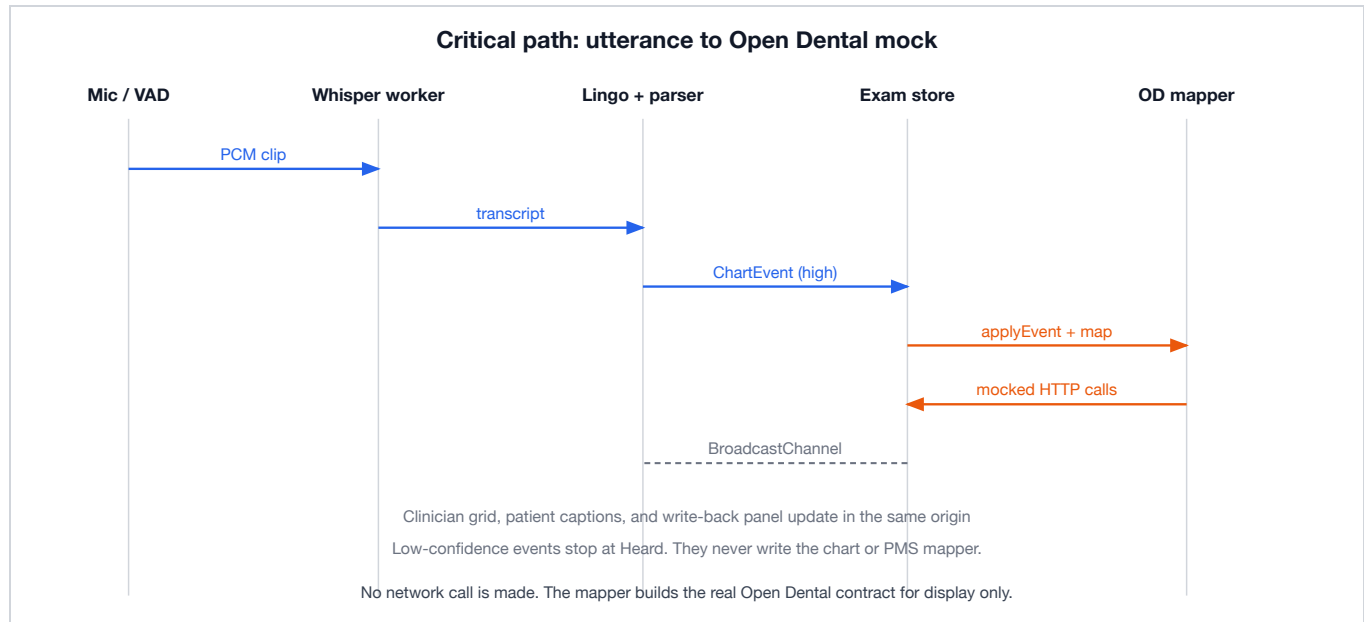


Figure 2. Critical path. Low-confidence work stops at Heard. High-confidence work updates the exam, then the mapper.

Sync versus async

Hop	Style	Contract
Mic to VAD	Async audio callback	Float32 frames, 4096-sample ScriptProcessor
VAD to Whisper	Async promise, 20 s timeout	PCM + sample rate
Transcript to parser	Sync	Raw string in, ChartEvent[] out
Parser to store	Sync, via bus publish that also loops local	ChartEvent
Store to PMS mapper	Sync, after applyEvent	Exam snapshot + event
Store to second tab	Async BroadcastChannel	chart-event messages
Future live Open Dental	HTTPS, not yet	Idempotent POST then PUT by measure id

Idempotency expectations (when the mock becomes live)

- Creating a perio exam is once per visit. Store the returned PerioExamNum.
- Creating a periomeasure is once per (exam, tooth, SequenceType). Duplicates should PUT.
- Procedurelogs for an existing crown or filling are once per (tooth, procCode) per session.
- Retries must not invent a second exam for the same chairside visit.

Mermaid (source of truth for the same flow)

```
sequenceDiagram
    participant Mic
    participant VAD
    participant Whisper
    participant Parser
    participant Store
    participant Mapper
    participant UI
    Mic->>VAD: PCM
    VAD->>Whisper: utterance clip
    Whisper->>Parser: transcript
    Parser->>Store: ChartEvent
    alt confidence low
        Store->>UI: Heard only
    else confidence high
        Store->>Store: applyEvent
        Store->>Mapper: exam + event
        Mapper->>Store: mocked HTTP calls
        Store->>UI: grid, captions, write-back
    end
end
```

7. Identity, auth, and secrets

AuthN today

None. Anyone with the laptop can open the clinician route and chart Andrew. That is correct for a hackathon demo and wrong for a clinic.

AuthZ today

None. There is one role: demo operator. Reset all is gated only by a confirm dialog.

Secrets today

Open Dental demo keys live in `.env.example` as obvious fakes (`DEMODEVKEY` / `DEMOCUSTOMKEY`). They are painted into the write-back panel so the Authorization header looks real. They must never be replaced with a clinic production pair in a committed file.

Target (not built)

Concern	Target
Clinic users	SSO against the practice IdP. No shared laptop password in the app.
Open Dental	Authorization: <code>ODFHIR {DeveloperKey}/{CustomerKey}</code> from a secrets manager, injected at runtime, never in git.
Browser	No long-lived PHI in <code>localStorage</code> . Session cookie or memory only.
Least privilege	Write periomeasures and procedurelogs for one <code>PatNum</code> . No patient search from the chairside client if a BFF can do it.

HIPAA, BAAs, and clinic SSO are explicitly out of scope for this build. Do not describe the current app as a covered entity system. The mapper is a contract rehearsal, not a compliance control.

8. Observability

Logs

Structured console.log prefixes are the current telemetry: [mic], [vad], [stt], [whisper-worker], [clinician]. They are for the operator's DevTools, not a SIEM.

Heard is the human-visible log of raw transcripts and confidence. The write-back panel is the human-visible log of PMS intent.

Metrics (what we would collect first in production)

- Utterance to chart latency (target under 1.5 s).
- Whisper empty / timeout / error rates.
- Share of transcripts marked unusable versus low-confidence versus written.
- Parser miss rate on rehearsal corpus (already unit-tested).
- When live: Open Dental 4xx/5xx, retry count, time to first PerioExamNum.

Traces

Not present. A future BFF should carry a visit id from utterance to PMS response. Do not invent a distributed tracer for a single tab.

Alerts

None. The operator watches Heard and the write-back panel. Production would page on PMS write failure after the hygienist leaves the room, not on a single low-confidence line.

9. Cost shape

Horizon	Order of magnitude	Notes
Demo laptop, as built	\$0 / month incremental	Local Vite, cached Whisper, no PMS traffic
Clinic pilot, still local STT	Workstation + IT time	Cost is people and Open Dental API access, not cloud inference
Hosted BFF + live OD	Low hundreds USD / month at one to three chairs	Tiny request volume. Do not buy a GPU fleet.

Top cost drivers later: (1) Open Dental developer program and clinic IT, not compute. (2) If STT moves to a server, GPU or reserved CPU for Whisper. (3) If a hosted model is used for patient copy, token spend. Keep that off the charting path.

Scaling cliffs: BroadcastChannel and in-tab Zustand will not survive ten chairs and a central server. That is a one-way door only if someone starts writing PHI into the SPA. The reversible path is a thin BFF that owns PMS writes while the parser stays local.

10. Failure modes and resilience

Failure	What the user sees	Blast radius	Recovery
No wifi / CDN cold start	Whisper stays loading	Live mic only	Warm model before the demo. Use rehearsal script.
WebGPU missing	Worker error or slow path	Live mic	Rehearsal. Later: WASM CPU fallback if we add it.
Wrong or locked iPhone mic	Heard empty, VAD never loud	Live mic	Unlock phone, one Listening tab, Refresh devices.
Junk transcript	Heard, amber, no chart write	None to PMS	Speak again or spacebar the script.
Low-confidence parse	Heard only	None to chart or PMS	Intended. Do not "helpfully" write.
Whisper timeout (20 s)	Heard shows the error	That utterance	Next utterance. Worker stays up unless dispose.
Tab refresh	Today's exam gone	Current visit	Re-seed from Andrew. Rehearse again. No RPO.
Second tab Listen	Mic mute / exclusive device	Audio	Only clinician Listen. Patient is display-only.
Future live OD 400/404	Would need an error row in write-back	That measure	Keep chart local. Retry PUT. Never delete the exam because one tooth failed.

RTO / RPO (demo): RTO is "hit Reset and play the script" (under a minute). RPO is zero durable today-exam by design. **Production targets (deferred):** RPO near zero for accepted high-confidence writes (local queue). RTO of a chair: another workstation can open the same visit from the BFF.

The rehearsal script is the primary resilience control. Treat it as production-grade for the demo, not as a test toy.

11. Migration and rollout

Phase 0 (current)

Browser-only. Mock Open Dental. Fictional Andrew. Safe to show in a room with wifi off after Whisper is cached.

Phase 1: contract lock

Keep the mapper. Add a feature flag that still does not send. Record fixtures from a sandbox Open Dental customer. Expand tests for PUT identity and SkipTooth.

Phase 2: outbound BFF

Introduce a small server the clinic owns. The browser posts ChartEvents or already-mapped PMS bodies to the BFF. The BFF holds ODFHIR keys and performs HTTPS. The SPA never sees production keys. This is the first one-way door we should take; it is still reversible if the BFF only writes sandbox PatNums.

Phase 3: live sandbox patient

One non-production PatNum. Hygienist confirms the write-back panel, then a human taps Send. No auto-send on every pocket. Chart remains source of truth if OD is down.

Phase 4: clinic pilot

SSO, visit identity, durable store, audit log, BAA review. Only then discuss auto-send. Dentrax or other PMS adapters plug in beside the Open Dental mapper, they do not replace the parser.

Do not migrate by "just pointing the current SPA at api.opendental.com." That would put clinic keys in a Vite bundle and write from an unmanaged laptop.

12. What implementation must respect

- No LLM on the path from transcript to tooth cell.
- Low confidence does not write PD, BOP, mobility, furcation, or PMS calls.
- Do not hardcode loopback URLs. Host, port, routes, model, and OD base stay in env.
- Do not start extra frontend or backend servers as part of this repo's normal demo. The operator already runs Vite.
- Parser and lingo stay function-based (includes, token lists). Do not regress to regex-led extraction.
- Open Dental field names stay canonical: MBvalue, Bvalue, DBvalue, MLvalue, Lvalue, DLvalue, SequenceType values as documented, -1 for unmeasured.
- BleedSupPlaqCalc uses flag sums (bleeding 1, suppuration 2, plaque 4, calculus 8). Today we only emit bleeding = 1.
- Existing restorations use procedurelogs with ProcStatus EC and ADA codes D2740 (crown) or D2391 (posterior composite), once per tooth and code.
- Reset tooth maps to SkipTooth (ToothValue 1), not a silent local-only clear if we are showing PMS intent.
- COOP same-origin and COEP credentialless stay on the Vite server. Removing them can break the worker.
- BroadcastChannel name chairside-events is part of the two-display contract.
- Andrew's March history is the comparison baseline. Do not "helpfully" seed today's pockets with March values.

13. Open questions

1. When a BFF exists, is the chairside client allowed to hold any PHI at rest, or only in memory for the visit?
2. Does the first live clinic want Open Dental Cloud API or an on-prem Open Dental API service?
The paths are the same; the hosting and firewall are not.
3. Should mobility and furcation also write commlog text, or only periomeasures?
4. Is there a clinic-standard recall write (POST /recalls Perio) at wrap-up, or is that front-desk work?
5. Do we restore a 3D hero tooth for #14, or is the photographed jaw the lasting patient surface?
6. Whisper model pin: stay on base.en, or offer small.en as an advanced setting after a measured latency test on the demo laptop?
7. Second PMS: is Dentrax Ascend a Phase 4 adapter, or a different product line?

None of these block the current demo. All of them block calling this a clinic system.

14. Appendix A: Open Dental call map

Public docs used: PerioExams, PerioMeasures, ProcedureLogs. Auth header: Authorization: ODFHIR {DeveloperKey}/{CustomerKey}.

Chairside event	Mocked call	Notes
First high-confidence write of the visit	POST /perioexams	PatNum, ExamDate, ProvNum, Note. 201. Stores PerioExamNum (demo starts at 31).
Pocket depths	POST /periomeasures then PUT /periomeasures/{id}	SequenceType Probing. Six surfaces, -1 if unknown.
Bleeding on probing	Same, SequenceType BleedSupPlaqCalc	1 on bleeding sites, 0 on measured quiet sites.
Recession	SequenceType GingMargin	Millimetres as surface values.
Mobility 0 to 3	SequenceType Mobility	ToothValue only. Surfaces -1.
Furcation 0 to 3	SequenceType Furcation	Usually Bvalue unless sites named.
Exam note ("no bleeding")	PUT /perioexams/{id}	Note overwrites. Create exam first if needed.
Reset tooth	SequenceType SkipTooth	ToothValue 1.
Note contains crown	POST /procedurelogs	procCode D2740, ProcStatus EC, ToothNum string.
Note contains filling or composite	POST /procedurelogs	procCode D2391, ProcStatus EC.

The clinician panel titled "Open Dental write-back" is the operator view of this map. It shows method, path, URL, headers, request JSON, and a mocked response. Nothing is transmitted.

15. Appendix B: As-built versus original BUILD spec

BUILD spec	As built	Decision
FastAPI + WebSocket + faster-whisper	Browser-only Whisper worker	Fewer moving parts
Patient Rita Shah	Patient Andrew, March 2026 history	Demo narrative change
3D hero tooth #14	Photographed upper and lower arches + meaning panel	Readable at distance; 3D remains optional later
Mock HTTP POST /mock/opendental/perio	Typed Open Dental REST shapes in the panel	Honest contract
Python web rtcvad / Silero	In-tab energy VAD	No native addon
Disk persistence of the visit	Memory only	Avoid leftover PHI on a shared laptop

16. Appendix C: Source map

Path	Role
src/audio/useLiveMic.ts	Capture loop, lingo, publish
src/audio/vad.ts	Energy VAD
src/audio/stt.ts / whisper.worker.ts	Whisper host and worker
src/audio/devices.ts	Mic pick (iPhone, Shokz)
src/domain/lingo.ts	Slang and commands
src/domain/parser.ts / clinicalParse.ts	ChartEvent extraction
src/domain/exam.ts	Apply events, colors, restorations
src/domain/translator.ts	Patient captions
src/store/examStore.ts	Visit state
src/pms/opendental.ts	PMS mapper
src/bus/channel.ts	BroadcastChannel
src/views/ClinicianView.tsx	Grid, script, write-back
src/views/PatientView.tsx	Jaw and meaning
src/views/WritebackDrawer.tsx	HTTP-shaped mock log
src/data/andrew.json	Seed
.env.example	Host, routes, Whisper, OD keys

MolarMind architecture pack. First issue, 12 September 2026. Companion to the in-app Architecture route. This document is as-built architecture, not a promise of clinic compliance.